

Q1.

- (a) Obtain the general solution of the differential equation

$$\tan x \frac{dy}{dx} + y = \sin x \tan x$$

where $0 < x < \frac{\pi}{2}$

(5)

- (b) Hence find the particular solution of this differential equation, given that $y = \frac{1}{2\sqrt{2}}$

when $x = \frac{\pi}{4}$

(2)

(Total 7 marks)

Q2.

- (a) Show that $\tan x$ is an integrating factor for the differential equation

$$\frac{dy}{dx} + \frac{\sec^2 x}{\tan x} y = \tan x$$

(2)

- (b) Hence solve this differential equation, given that $y = 3$ when $x = \frac{\pi}{4}$.

(6)

(Total 8 marks)

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Q3.

- (a) Find the values of the constants a , b and c for which $a + bx + cxe^{-3x}$ is a particular integral of the differential equation

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 3y = 3x - 8e^{-3x}$$

(5)

- (b) Hence find the general solution of this differential equation.

(3)

- (c) Hence express y in terms of x , given that $y = 1$ when $x = 0$ and that $\frac{dy}{dx} \rightarrow -1$ as $x \rightarrow \infty$.

(4)

(Total 12 marks)

Q4.

- (a) Differentiate $\ln(\ln x)$ with respect to x .

(1)

- (b) (i) Show that $\ln x$ is an integrating factor for the first-order differential equation

$$\frac{dy}{dx} + \frac{1}{x \ln x} y = 9x^2, \quad x > 1$$

(2)

- (ii) Hence find the solution of this differential equation, given that $y = 4e^3$ when $x = e$.

(6)

(Total 9 marks)

Q5.

- (a) By using an integrating factor, find the general solution of the differential equation

$$\frac{dy}{dx} + \frac{2}{x} y = \ln x$$

(7)

- (b) Hence, given that $y \rightarrow 0$ as $x \rightarrow 0$, find the value of y when $x = 1$.

(3)

(Total 10 marks)

Q6.

- (a) By using an integrating factor, find the general solution of the differential equation

$$\frac{dy}{dx} + \frac{4}{2x+1} y = 4(2x+1)^5$$

giving your answer in the form $y = f(x)$.

(7)

- (b) The gradient of a curve at any point (x, y) on the curve is given by the differential equation

$$\frac{dy}{dx} = 4(2x+1)^5 - \frac{4}{2x+1} y$$

The point whose x -coordinate is zero is a stationary point of the curve. Using your answer to part (a), find the equation of the curve.

(3)

(Total 10 marks)

Q7.

- (a) Find the values of the constants p and q for which $p \sin x + q \cos x$ is a particular integral of the differential equation

$$\frac{dy}{dx} + 5y = 13 \cos x$$

(3)

- (b) Hence find the general solution of this differential equation.

(3)

(Total 6 marks)

Q8.

- (a) Find the values of the constants p and q for which $p + qxe^{-2x}$ is a particular integral of the differential equation

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 2y = 4 - 9e^{-2x}$$

(5)

- (b) Hence find the general solution of this differential equation.

(3)

- (c) Hence express y in terms of x , given that $y = 4$ when $x = 0$ and that $\frac{dy}{dx} \rightarrow 0$ as $x \rightarrow \infty$.

(4)

(Total 12 marks)

Q9.

By using an integrating factor, find the solution of the differential equation

$$\frac{dy}{dx} + (\cot x)y = \sin 2x, \quad 0 < x < \frac{\pi}{2}$$

given that $y = \frac{1}{2}$ when $x = \frac{\pi}{6}$

(Total 10 marks)

Q10.

It is given that y satisfies the differential equation

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 2e^{-2x}$$

- (a) Find the value of the constant p for which $y = pxe^{-2x}$ is a particular integral of the given differential equation.

(4)

- (b) Solve the differential equation, expressing y in terms of x , given that $y = 2$ and $\frac{dy}{dx} = 0$ when $x = 0$.

(8)
(Total 12 marks)

Q11.

Find the general solution of the differential equation

$$\frac{d^2 y}{dx^2} + 4y = 8x^2 + 9 \sin x$$

(Total 8 marks)

Q12.

- (a) Find the value of the constant k for which $k \sin 2x$ is a particular integral of the differential equation

$$\frac{d^2 y}{dx^2} + y = \sin 2x$$

(3)

- (b) Hence find the general solution of this differential equation.

(4)

(Total 7 marks)

Q13.

By using an integrating factor, find the solution of the differential equation

$$\frac{dy}{dx} + \frac{3}{x}y = (x^4 + 3)^{\frac{3}{2}}$$

given that $y = \frac{1}{5}$ when $x = 1$.

(Total 9 marks)

Q14.

- (a) Show that $\frac{1}{x^2}$ is an integrating factor for the first-order differential equation

$$\frac{dy}{dx} - \frac{2}{x}y = x$$

(3)

- (b) Hence find the general solution of this differential equation, giving your answer in the form $y = f(x)$.

(4)

(Total 7 marks)

Q15.

By using an integrating factor, find the solution of the differential equation

$$\frac{dy}{dx} - y \tan x = 2 \sin x$$

given that $y = 2$ when $x = 0$.

(Total 9 marks)

Q16.

It is given that y satisfies the differential equation

$$\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + 5y = 8 \sin x + 4 \cos x$$

- (a) Find the value of the constant k for which $y = k \sin x$ is a particular integral of the given differential equation.

(3)

- (b) Solve the differential equation, expressing y in terms of x , given that $y = 1$ and $\frac{dy}{dx} = 4$ when $x = 0$.

(8)

(Total 11 marks)

Q17.

- (a) Find the general solution of the differential equation

$$\frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} + 5y = 5$$

(6)

- (b) Hence express y in terms of x , given that $y = 2$ and $\frac{dy}{dx} = 3$ when $x = 0$.

(4)

(Total 10 marks)

Q18.

By using an integrating factor, find the solution of the differential equation

$$\frac{dy}{dx} + \frac{4x}{x^2 + 1} y = x$$

given that $y = 1$ when $x = 0$. Give your answer in the form $y = f(x)$.

(Total 9 marks)

Q19.

- (a) Find the values of the constants a , b , c and d for which $a + bx + c \sin x + d \cos x$ is

a particular integral of the differential equation

$$\frac{dy}{dx} - 3y = 10 \sin x - 3x$$

(4)

- (b) Hence find the general solution of this differential equation.

(3)

(Total 7 marks)

Q20.

- (a) A differential equation is given by

$$x \frac{d^2 y}{dx^2} - \frac{dy}{dx} = 3x^2$$

Show that the substitution

$$u = \frac{dy}{dx}$$

transforms this differential equation into

$$\frac{du}{dx} - \frac{1}{x}u = 3x$$

(2)

- (b) By using an integrating factor, find the general solution of

$$\frac{du}{dx} - \frac{1}{x}u = 3x$$

giving your answer in the form $u = f(x)$.

(6)

- (c) Hence find the general solution of the differential equation

$$x \frac{d^2 y}{dx^2} - \frac{dy}{dx} = 3x^2$$

giving your answer in the form $y = g(x)$.

(2)

(Total 10 marks)

Q21.

- (a) Find the general solution of the differential equation

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} - 3y = 10e^{-2x} - 9$$

(10)

- (b) Hence express y in terms of x , given that $y = 7$ when $x = 0$ and that $\frac{dy}{dx} \rightarrow 0$ as $x \rightarrow \infty$.

(4)

(Total 14 marks)

Q22.

- (a) Show that x^2 is an integrating factor for the first-order differential equation

$$\frac{dy}{dx} + \frac{2}{x}y = 3(x^3 + 1)^{\frac{1}{2}}$$

(3)

- (b) Solve this differential equation, given that $y = 1$ when $x = 2$.

(6)

(Total 9 marks)

Q23.

Find the general solution of the differential equation

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 3y = 6 + 5\sin x$$

(Total 12 marks)

Q24.

- (a) Find the value of the constant k for which kx^2e^{5x} is a particular integral of the differential equation

$$\frac{d^2y}{dx^2} - 10\frac{dy}{dx} + 25y = 6e^{5x}$$

(6)

- (b) Hence find the general solution of this differential equation.

(4)

(Total 10 marks)

Q25.

By using an integrating factor, find the solution of the differential equation

$$\frac{dy}{dx} + (\tan x)y = \sec x$$

given that $y = 3$ when $x = 0$.

(Total 8 marks)

Q26.

- (a) Find the roots of the equation $m^2 + 2m + 2 = 0$ in the form $a + ib$.

(2)

- (b) (i) Find the general solution of the differential equation

$$\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + 2y = 4x$$

(6)

- (ii) Hence express y in terms of x , given that $y = 1$ and $\frac{dy}{dx} = 2$ when $x = 0$.

(4)

(Total 12 marks)

Q27.

- (a) Show that $y = x^3 - x$ is a particular integral of the differential equation

$$\frac{dy}{dx} + \frac{2xy}{x^2 - 1} = 5x^2 - 1$$

(3)

- (b) By differentiating $(x^2 - 1)y = c$ implicitly, where y is a function of x and c is a

constant, show that $y = \frac{c}{x^2 - 1}$ is a solution of the differential equation

$$\frac{dy}{dx} + \frac{2xy}{x^2 - 1} = 0$$

(3)

- (c) Hence find the general solution of

$$\frac{dy}{dx} + \frac{2xy}{x^2 - 1} = 5x^2 - 1$$

(2)

(Total 8 marks)

Q28.

- (a) The function $y(x)$ satisfies the differential equation

$$\frac{dy}{dx} = f(x, y)$$

where

$$f(x, y) = x \ln x + \frac{y}{x}$$

and

$$y(1) = 1$$

- (i) Use the Euler formula

$$y_{r+1} = y_r + h f(x_r, y_r)$$

with $h = 0.1$, to obtain an approximation to $y(1.1)$.

(3)

- (ii) Use the formula

$$y_{r+1} = y_{r-1} + 2h f(x_r, y_r)$$

with your answer to part (a)(i) to obtain an approximation to $y(1.2)$, giving your answer to three decimal places.

(4)

- (b) (i) Show that $\frac{1}{x}$ is an integrating factor for the first-order differential equation

$$\frac{dy}{dx} - \frac{1}{x}y = x \ln x$$

(3)

- (ii) Solve this differential equation, given that $y = 1$ when $x = 1$.

(6)

- (iii) Calculate the value of y when $x = 1.2$, giving your answer to three decimal places.

(1)

(Total 17 marks)

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Q29.

It is given that y satisfies the differential equation

$$\frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} + 4y = 8x - 10 \cos 2x$$

- (a) Show that $y = 2x + \sin 2x$ is a particular integral of the given differential equation.

(3)

- (b) Find the general solution of the differential equation.

(4)

- (c) Hence express y in terms of x , given that $y = 2$ and $\frac{dy}{dx} = 0$ when $x = 0$.

(4)

(Total 11 marks)

Q30.

- (a) Show that $\sin x$ is an integrating factor for the differential equation

$$\frac{dy}{dx} + (\cot x)y = 2 \cos x$$

(3)

- (b) Solve this differential equation, given that $y = 2$ when $x = \frac{\pi}{2}$.

(6)

(Total 9 marks)

Q31.

- (a) Show that the substitution

$$u = \frac{dy}{dx} + 2y$$

transforms the differential equation

$$\frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} + 4y = e^{-2x}$$

into

$$\frac{du}{dx} + 2u = e^{-2x}$$

(4)

- (b) By using an integrating factor, or otherwise, find the general solution of

$$\frac{du}{dx} + 2u = e^{-2x}$$

giving your answer in the form $u = f(x)$.

(5)

- (c) Hence find the general solution of the differential equation

$$\frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} + 4y = e^{-2x}$$

giving your answer in the form $y = g(x)$.

(5)

(Total 14 marks)

Q32.

During the design of an aircraft, part of its tail section is subjected to vibrations. At time t , the displacement of the tail section, x , is modelled by the differential equation

$$\ddot{x} + 121x = 3 \sin \omega t.$$

Different values of ω are used to simulate different flight conditions.

Find the general solution of this differential equation when $\omega \neq 11$.

(Total 6 marks)



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